Inference for logistic regression models

Recap: hypothesis testing

Model:

$$Admit_i \sim Bernoulli(\pi_i)$$

$$\log\left(\frac{\pi_i}{1-\pi_i}\right) = \beta_0 + \beta_1 GPA_i$$

Research question: is there a relationship between GPA and acceptance to a graduate program?

Hypotheses: $H_0 : \beta_1 = 0 \qquad H_A : \beta_1 \neq 0$

Option 1: Wald test (aka z-test)

```
m1 <- glm(admit ~ gpa, family = binomial, data = grad
summary(m1)$coefficients
```

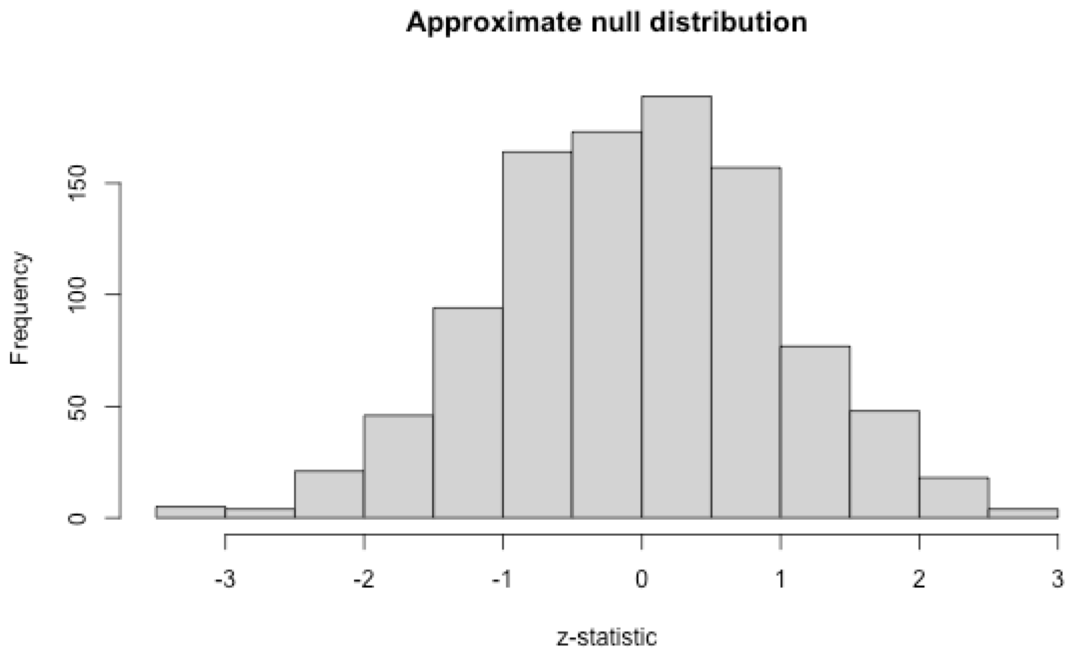
```
##              Estimate Std. Error   z value     Pr(>|z|)
## (Intercept) -4.357587  1.0353170 -4.20894 2.565715e-05
## gpa          1.051109  0.2988694  3.51695 4.365354e-04
```

$$z = \frac{\widehat{\beta}_1}{SE(\widehat{\beta}_1)} = 3.517$$

> Is this an "unusual" z-statistic, if H_0 is true?

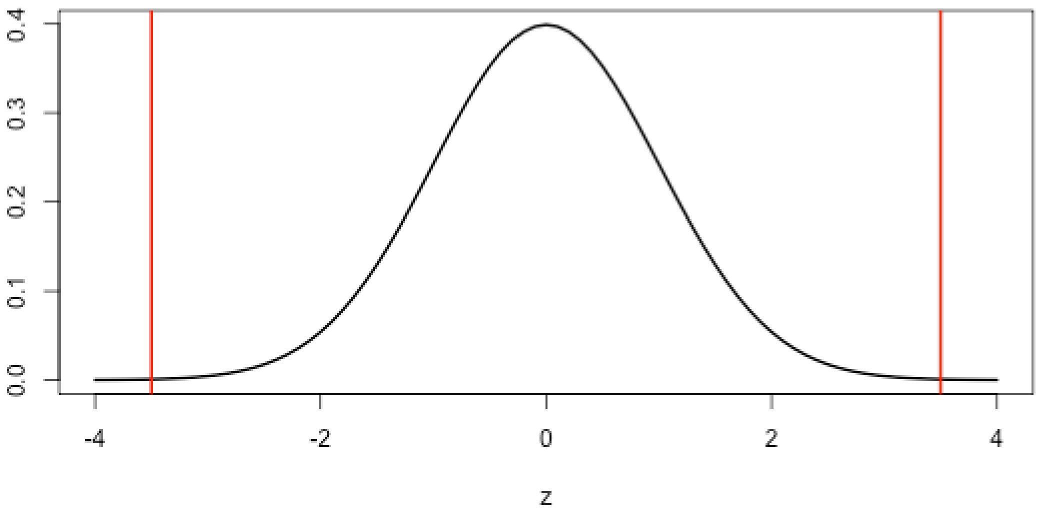
Option 1: Wald test (aka z-test)

Approximate null distribution:



Option 1: Wald test (aka z-test)

Observed test statistic: $z = 3.517$




```r
2*pnorm(abs(3.517), lower.tail=F)
```

```
## [1] 0.0004364538
```

Option 1: Wald test (aka z-test)

```
2*pnorm(abs(3.517), lower.tail=F)
```

```
## [1] 0.0004364538
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```
summary(m1)$coefficients
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```

Recap: hypothesis testing

Full model:

$$Admit_i \sim Bernoulli(\pi_i)$$

$$\log\left(\frac{\pi_i}{1 - \pi_i}\right) = \beta_0 + \beta_1 GPA_i$$

Wish to test $H_0 : \beta_1 = 0$ vs. $H_A : \beta_1 \neq 0$.

> What is my **reduced model** for these hypotheses?

Recap: hypothesis testing

Full model:

$$Admit_i \sim Bernoulli(\pi_i)$$

$$\log\left(\frac{\pi_i}{1-\pi_i}\right) = \beta_0 + \beta_1 GPA_i$$

Wish to test $H_0 : \beta_1 = 0$ vs. $H_A : \beta_1 \neq 0$.

Reduced model:

$$\log\left(\frac{\pi_i}{1-\pi_i}\right) = \beta_0$$

Activity

Work on the activity (handout), then we will discuss as a class.

Activity

Fitted reduced model:

```
glm(admit ~ 1, family = binomial, data = grad_app)
```

```
##
## Call:  glm(formula = admit ~ 1, family = binomial, data
##
## Coefficients:
## (Intercept)
##     -0.7653
##
## Degrees of Freedom: 399 Total (i.e. Null);  399 Residual
## Null Deviance:          500
## Residual Deviance: 500      AIC: 502
```

> What do you notice about the "Null" and "Residual" deviance?

Activity

```
glm(admit ~ gpa, family = binomial, data = grad_app)
```

```
##
## Call:  glm(formula = admit ~ gpa, family = binomial, dat
##
## Coefficients:
## (Intercept)          gpa
##      -4.358        1.051
##
## Degrees of Freedom: 399 Total (i.e. Null);  398 Residual
## Null Deviance:          500
## Residual Deviance: 487      AIC: 491
```

What is the drop-in-deviance statistic for the test $H_0 : \beta_1 = 0$ vs. $H_A : \beta_1 \neq 0$?

Activity

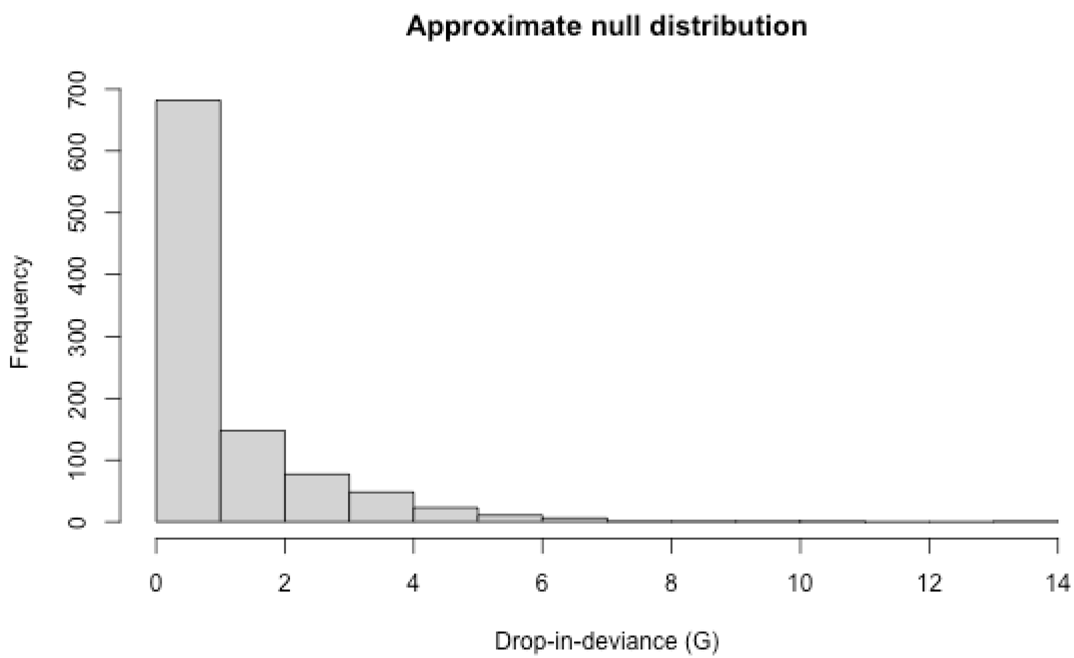
```
##
## Call:  glm(formula = admit ~ gpa, family = binomial, dat
##
## Coefficients:
## (Intercept)          gpa
##      -4.358        1.051
##
## Degrees of Freedom: 399 Total (i.e. Null);  398 Residual
## Null Deviance:          500
## Residual Deviance: 487      AIC: 491
```

$$G = \text{deviance}_{\text{reduced}} - \text{deviance}_{\text{full}} = 13$$

> How do we know if this test statistic is "unusual" under the null hypothesis?

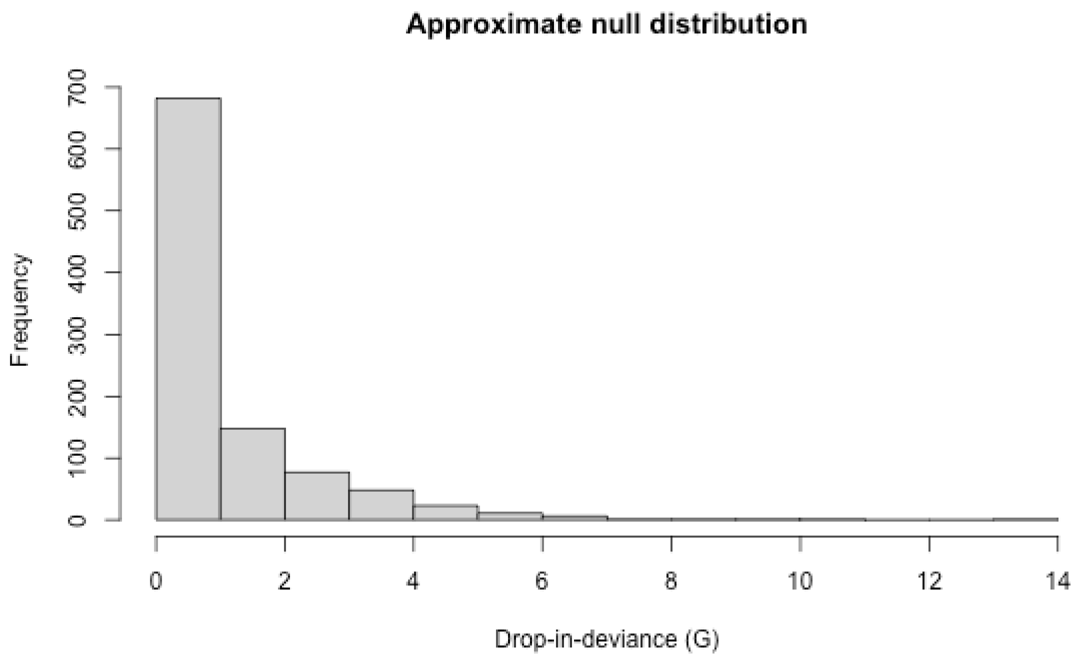
Activity

Approximate null distribution:



> Our observed test statistic is $G = 13$. Do you think that the null hypothesis is true?

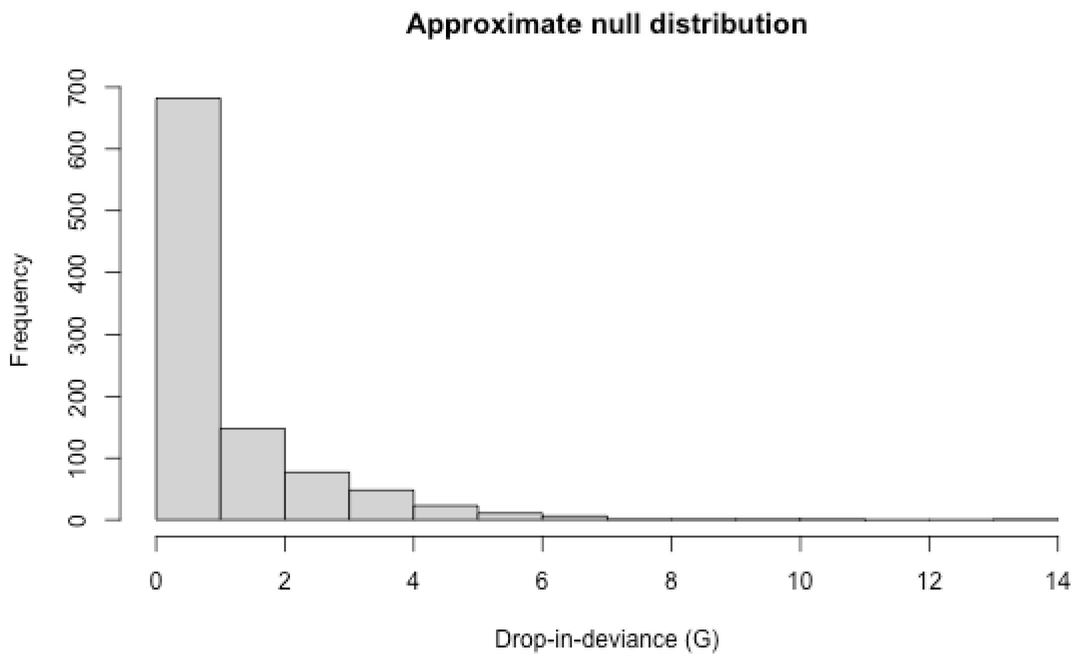
Calculating a p-value



p-value = $P(G > \text{observed value} \mid H_0 \text{ true})$

Need a distribution to calculate p-value!

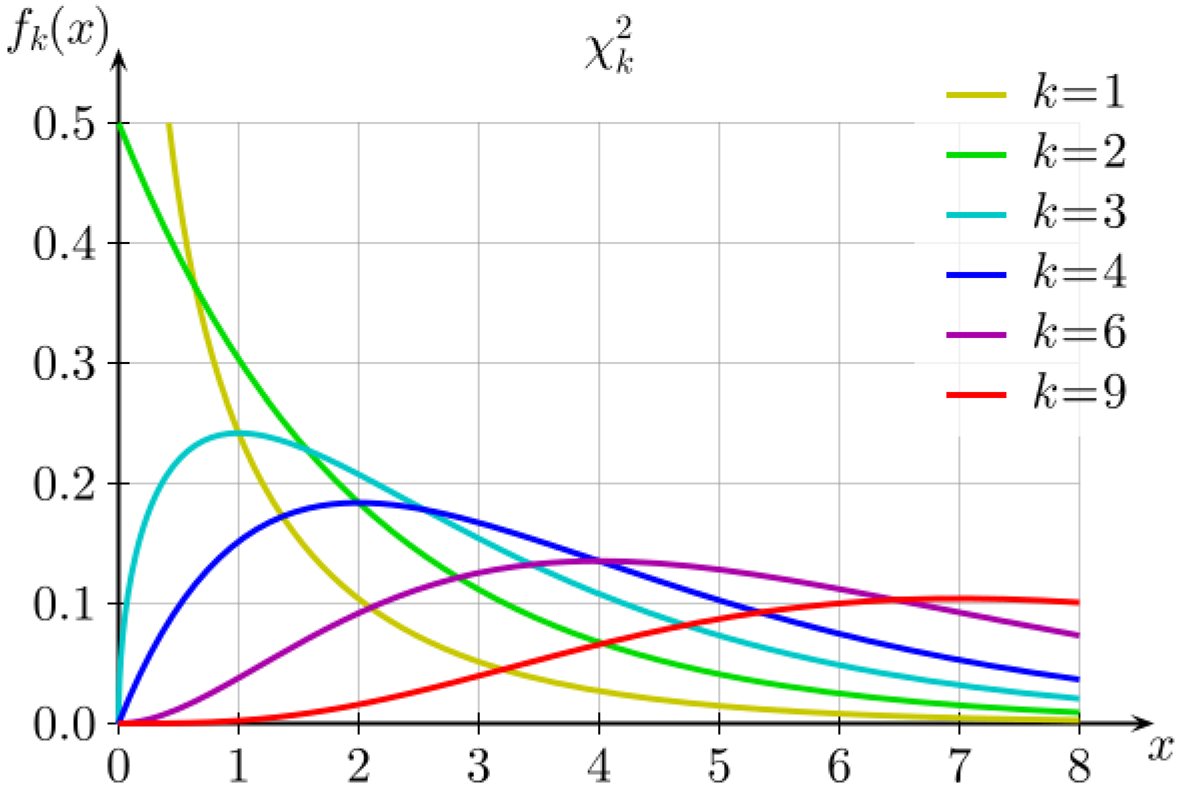
Class activity



Why is the drop-in-deviance always non-negative?

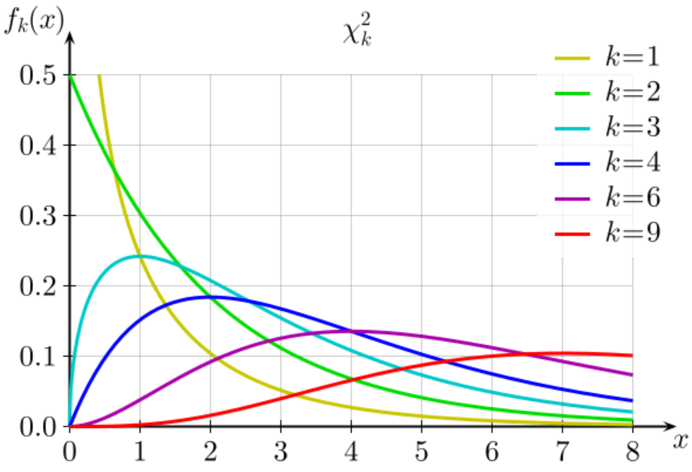
Class activity

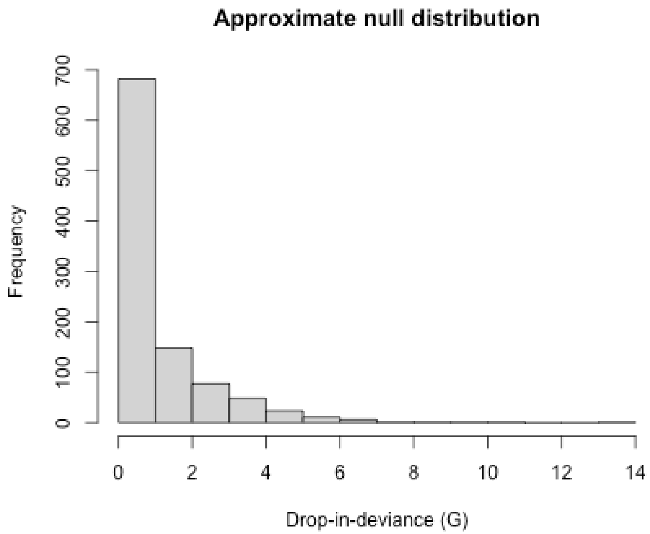
Under $H_0, G \sim \chi^2_k$



Class activity

$$H_0 : \beta_1 = 0 \qquad H_A : \beta_1 \neq 0$$





Under H_0, $G \sim \chi^2_k$. Which value of k seems appropriate for the null distribution?

Class activity

- Mean of χ^2_k is k
- Variance of χ^2_k is $2k$

```
mean(null_statistics)
```

```
## [1] 1.020683
```

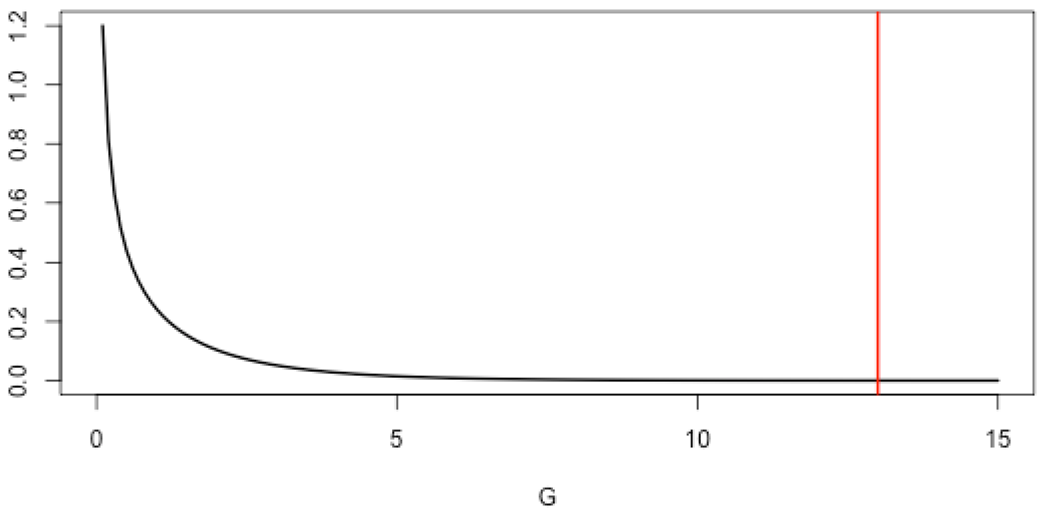
```
var(null_statistics)
```

```
## [1] 2.00598
```

Calculating a p-value

$H_0: \beta_1 = 0 \qquad H_A: \beta_1 \neq 0 \qquad G = \text{deviance}_{\text{reduced}} - \text{deviance}_{\text{full}} = 13$

Under $H_0, G \sim \chi^2_1$




```
pchisq(13, df=1, lower.tail=F)
```

```
## [1] 0.000311491
```

Drop-in-deviance test (aka likelihood ratio test)

Goal: Compare nested models (full and reduced models)

Steps:

- Calculate deviance for full and reduced models
- Test statistic: $G = \text{deviance}_{\text{reduced}} - \text{deviance}_{\text{full}}$
- Null distribution: under $H_0, G \sim \chi^2_q$

Wald test vs. drop-in-deviance test

Wald test (z-test)

- like t-tests
- test a *single* parameter
- some example hypotheses:
  - $H_0 : \beta_1 = 0$ vs. $H_A : \beta_1 \neq 0$
  - $H_0 : \beta_1 = 0$ vs. $H_A : \beta_1 > 0$
- null distribution: $N(0, 1)$

Drop-in-deviance test (likelihood ratio test)

- like nested F-tests
- test one or more parameters
- some example hypotheses:
  - $H_0 : \beta_1 = 0$ vs. $H_A : \beta_1 \neq 0$
  - $H_0 : \beta_1 = \beta_2 = 0$ vs H_A: at least one of $\beta_1, \beta_2 \neq 0$
- null distribution: χ^2_q

Activity

Work on the activity (handout), then we will discuss as a class.

Activity

$$Admit_i \sim Bernoulli(\pi_i)$$

$$\log\left(\frac{\pi_i}{1-\pi_i}\right) = \beta_0 + \beta_1 GPA_i + \beta_2 GRE_i + \beta_3 Rank2_i + \beta_4 Rank3_i + \beta_5 Rank4_i$$

> We want to test whether there is a relationship between Rank and grad school acceptance, after accounting for GPA and GRE. What are my null and alternative hypotheses?

Activity

$$Admit_i \sim Bernoulli(\pi_i)$$

$$\log\left(\frac{\pi_i}{1-\pi_i}\right) = \beta_0 + \beta_1 GPA_i + \beta_2 GRE_i + \beta_3 Rank2_i + \beta_4 Rank3_i + \beta_5 Rank4_i$$

$H_0 : \beta_3 = \beta_4 = \beta_5 = 0$

H_A : at least one of $\beta_3, \beta_4, \beta_5 \neq 0$

> What is the reduced model?

Activity

```
##
## Call:  glm(formula = admit ~ gpa + gre + factor(rank), family = binomial,
##     data = grad_app)
##
## Coefficients:
##   (Intercept)            gpa            gre  factor(rank)2  factor(rank)3
##     -3.989979       0.804038       0.002264      -0.675443      -1.340204
## factor(rank)4
##     -1.551464
##
## Degrees of Freedom: 399 Total (i.e. Null);  394 Residual
## Null Deviance:        500
## Residual Deviance: 458.5     AIC: 470.5
```

```
##
## Call:  glm(formula = admit ~ gpa + gre, family = binomial, data = grad_app
##
## Coefficients:
## (Intercept)          gpa          gre
##   -4.949378     0.754687     0.002691
##
## Degrees of Freedom: 399 Total (i.e. Null);  397 Residual
## Null Deviance:        500
## Residual Deviance: 480.3     AIC: 486.3
```

Activity

$H_0 : \beta_3 = \beta_4 = \beta_5 = 0$

H_A : at least one of $\beta_3, \beta_4, \beta_5 \neq 0$

Drop-in-deviance test: $G = 21.8$

> What is the distribution of G under the null hypothesis?

Activity

$H_0 : \beta_3 = \beta_4 = \beta_5 = 0$

H_A : at least one of $\beta_3, \beta_4, \beta_5 \neq 0$

Drop-in-deviance test: $G = 21.8$

Under $H_0, G \sim \chi^2_3$

```
pchisq(21.8, df=3, lower.tail=F)
```

```
## [1] 7.178958e-05
```